

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 2**

**COURSE CODE:** CSA51 **COURSE NAME:** Cryptography and Network Security

---

### **COURSE OUTCOME COVERED**

**CO4:** *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

*Assessment Tool Weightage: 20%*

---

**QUESTIONS: 5**

**TOTAL MARKS: 25**

### **Annexure A**

#### ***Industry Problem-Solving Task***

---

#### ***Code of Conduct:***

*I S. Harsha Vardhan Reddy(192325029) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***  
*S. Harsha Vardhan Reddy*

## **Annexure A**

### **1. SSL/TLS Overhead and Web Server Performance Optimization**

A web server uses TLS 1.3 for secure communication. Each HTTP response carries 2000 bytes of payload. TLS adds 25 bytes (record header + authentication tag) and handshake overhead of 3 KB per session. The server handles 10,000 requests per second, with 1,000 new TLS handshakes per second.

#### **Task:**

- A. Calculate the total data transmitted per second including TLS overhead.
  - B. Determine the percentage overhead due to TLS encryption.
  - C. Estimate bandwidth consumed due to handshake overhead per second.
  - D. Analyze the trade-off between security and latency, and propose optimization techniques (e.g., session resumption, HTTP/2, or QUIC).
- 

### **2. PGP-Based Email Encryption Load in Enterprise System**

An enterprise email server encrypts emails using PGP. Each email contains a 10 MB attachment. PGP introduces 800 bytes overhead for key encryption and signatures. The system processes 5,000 emails per day.

#### **Task:**

- A. Calculate the total data transmitted per day including overhead.
  - B. Determine the percentage increase in storage due to PGP.
  - C. Analyze the computational overhead during encryption/decryption.
  - D. Recommend optimization strategies (e.g., compression, batching, or hardware acceleration).
- 

### **3. IPSec Transport vs Tunnel Mode Performance Comparison**

A company compares IPSec transport mode and tunnel mode for securing internal communications. Original packet size = 1000 bytes payload + 20-byte header. Transport mode adds 30 bytes, while tunnel mode adds 50 bytes overhead. Traffic volume = 20,000 packets per second.

#### **Task:**

- A. Compute total packet size for both modes.
  - B. Calculate bandwidth usage for each mode per second.
  - C. Compare efficiency and security benefits of both modes.
  - D. Recommend the appropriate mode for internal vs external communication with justification.
-

#### **4. S/MIME Certificate Management in Large Organization**

An organization manages 12,000 employees using S/MIME certificates. Each certificate is 2 KB, and certificate renewal happens every year. During renewal, both old and new certificates are stored simultaneously for 1 month.

**Task:**

- A. Calculate total storage required during the renewal period.
  - B. Estimate certificate distribution overhead across the network.
  - C. Analyze challenges in certificate lifecycle management at scale.
  - D. Propose optimization strategies (e.g., automation, centralized PKI, or short-lived certificates).
- 

#### **5. Email Encryption Latency in Cloud-Based Systems**

A cloud email service encrypts outgoing emails using AES (data) and RSA (key exchange). AES encryption takes 0.2 ms per MB, and RSA key exchange takes 5 ms per email. Each email has 8 MB attachment. The system processes 8,000 emails per hour.

**Task:**

- A. Calculate total encryption time per email.
- B. Estimate total processing time per hour.
- C. Analyze the performance bottleneck between symmetric and asymmetric operations.
- D. Recommend optimization strategies (e.g., ECC adoption, parallel processing, or hardware acceleration).

### RUBRICS FOR CALCULATION:

| Criteria                            | Weightage | Level 4 – Exemplary                                                          | Level 3 – Proficient                                 | Level 2 – Developing                           | Level 1 – Beginning                                |
|-------------------------------------|-----------|------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|----------------------------------------------------|
| Understanding of Industry Problem   | 20%       | Comprehensive understanding of real-world problem and constraints            | Good understanding with minor gaps in requirements   | Basic understanding; some requirements unclear | Poor understanding ; misinterprets problem context |
| Application of Engineering Concepts | 25%       | Applies advanced and relevant concepts effectively to solve the problem      | Applies appropriate concepts with minor errors       | Limited application of concepts                | Incorrect or no application of concepts            |
| Solution Design                     | 25%       | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps        | Basic solution with limited feasibility        | Unclear or impractical solution                    |
| Use of Tools                        | 15%       | Effective use of modern tools, technologies, and real data                   | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools           | Minimal or incorrect use of tools                  |
| Analysis                            | 10%       | Thorough analysis with validated results and performance evaluation          | Good analysis with reasonable validation             | Basic analysis with limited validation         | Poor or no analysis                                |
| Professionalism                     | 5%        | Highly professional, well-documented, and clearly presented work             | Good documentation and presentation                  | Basic documentation with some gaps             | Poor documentation and presentation                |

## **1. SSL/TLS Overhead and Web Server Performance Optimization**

Given: HTTP payload per response = 2000 bytes; TLS overhead = 25 bytes; requests = 10,000 requests/second; new TLS handshakes = 1,000/second; handshake overhead = 3 KB/session.

### **A. Total data transmitted per second**

Payload per second =  $2000 \times 10,000 = 20,000,000$  bytes/sec.

TLS record overhead =  $25 \times 10,000 = 250,000$  bytes/sec.

Data including TLS record overhead =  $20,000,000 + 250,000 = 20,250,000$  bytes/sec = 20.25 MB/s.

Handshake overhead =  $3 \text{ KB} \times 1,000 = 3,000 \text{ KB/s} = 3 \text{ MB/s}$ .

Total including handshake overhead =  $20.25 + 3 = 23.25 \text{ MB/s}$ .

Answer: 23.25 MB/s.

### **B. Percentage overhead due to TLS encryption**

Record overhead percentage =  $(25 / 2000) \times 100 = 1.25\%$ .

If handshake overhead is also included:  $((250,000 + 3,000,000) / 20,000,000) \times 100 = 16.25\%$ .

Answer: TLS record overhead is 1.25% per response; total TLS overhead including handshakes is 16.25% relative to the payload.

### **C. Bandwidth consumed due to handshake overhead**

$3 \text{ KB} \times 1,000 = 3,000 \text{ KB/s} = 3 \text{ MB/s}$ .

Answer: 3 MB/s.

### **D. Security, latency, and optimization**

TLS provides confidentiality, integrity, and authentication, but handshakes introduce additional CPU usage, network traffic, and latency.

TLS session resumption: avoids a complete handshake for returning clients.

HTTP/2: multiplexes multiple requests over a single TLS connection.

Connection reuse/keep-alive: reduces repeated handshakes.

QUIC/HTTP/3: reduces connection-establishment latency and handles network changes efficiently.

Hardware acceleration: reduces cryptographic CPU overhead.

Conclusion: TLS should be retained for security, while session resumption and connection reuse can significantly reduce its performance impact.

## **2. PGP-Based Email Encryption Load in Enterprise System**

Given: attachment = 10 MB/email; PGP overhead = 800 bytes/email; emails = 5,000/day.

### **A. Total data transmitted per day**

Total attachment data =  $10 \text{ MB} \times 5,000 = 50,000 \text{ MB} = 50 \text{ GB}$ .

PGP overhead =  $800 \times 5,000 = 4,000,000$  bytes  $\approx 4$  MB.

Total =  $50,000 \text{ MB} + 4 \text{ MB} = 50,004 \text{ MB} \approx 50.004 \text{ GB/day}$ .

Answer: 50.004 GB/day.

### **B. Percentage increase in storage**

Using 10 MB = 10,000,000 bytes:  $(800 / 10,000,000) \times 100 = 0.008\%$ .

Answer: 0.008% storage increase per email based on the given figures.

### **C. Computational overhead**

PGP normally uses asymmetric cryptography for key-related operations and digital signatures, while symmetric encryption handles the large attachment efficiently. Public-key operations are computationally expensive, and encryption/decryption consumes CPU resources. At 5,000 emails/day, the overhead can become significant if server CPU capacity is limited or many emails are processed simultaneously.

### **D. Optimization strategies**

Compression: compress attachments before encryption to reduce data size.

Batch processing: process encryption jobs efficiently in groups.

Hardware acceleration: use cryptographic acceleration where available.

Efficient algorithms: use modern symmetric encryption for bulk data.

Parallel processing: distribute encryption/decryption across multiple CPU cores.

Conclusion: Storage overhead is very small, but computational overhead can be significant, so CPU optimization and parallel processing are important.

## **3. IPSec Transport vs. Tunnel Mode Performance Comparison**

Given: payload = 1000 bytes; original header = 20 bytes; transport-mode overhead = 30 bytes; tunnel-mode overhead = 50 bytes; traffic = 20,000 packets/second.

### **A. Total packet size**

Original packet =  $1000 + 20 = 1020$  bytes.

Transport mode =  $1020 + 30 = 1050$  bytes.

Tunnel mode =  $1020 + 50 = 1070$  bytes.

### **B. Bandwidth usage**

Transport mode:  $1050 \times 20,000 = 21,000,000$  bytes/sec = 21 MB/s.

Tunnel mode:  $1070 \times 20,000 = 21,400,000$  bytes/sec = 21.4 MB/s.

| Mode      | Packet Size | Bandwidth |
|-----------|-------------|-----------|
| Transport | 1050 bytes  | 21 MB/s   |
| Tunnel    | 1070 bytes  | 21.4 MB/s |

### **C. Efficiency and security**

Transport mode has lower overhead and better bandwidth efficiency. It protects the IP payload and is suitable when communicating directly between trusted hosts.

Tunnel mode has higher overhead. It encrypts the complete original IP packet and encapsulates it in a new packet, providing better protection of the original packet structure and making it suitable for VPNs and communication across untrusted networks.

Transport efficiency =  $(1000 / 1050) \times 100 = 95.24\%$ .

Tunnel efficiency =  $(1000 / 1070) \times 100 = 93.46\%$ .

#### **D. Recommendation**

Internal communication: Use IPSec transport mode when hosts are trusted and lower overhead is important.

External communication/VPN: Use IPSec tunnel mode because it provides stronger network-level protection and hides the original packet structure.

#### **4. S/MIME Certificate Management in Large Organization**

Given: employees = 12,000; certificate size = 2 KB; renewal = once/year; old and new certificates are stored simultaneously for one month.

##### **A. Total storage during renewal**

One set of certificates =  $12,000 \times 2 \text{ KB} = 24,000 \text{ KB} = 24 \text{ MB}$ .

During renewal, two certificates are stored:  $24 \text{ MB} \times 2 = 48 \text{ MB}$ .

Answer: 48 MB.

##### **B. Certificate distribution overhead**

Distributing one complete set =  $12,000 \times 2 \text{ KB} = 24 \text{ MB}$ .

If both old and new certificates are distributed =  $24 \text{ MB} + 24 \text{ MB} = 48 \text{ MB}$ .

The exact network overhead depends on distribution attempts, retransmissions, and protocol overhead.

##### **C. Lifecycle management challenges**

Certificate issuance and identity verification.

Certificate renewal and expiration management.

Certificate revocation.

Lost or compromised private keys.

Employee joining and leaving.

Certificate storage and backup.

Email-client compatibility.

Monitoring expiration dates.

Failure to renew certificates on time can cause users to lose secure email communication.

##### **D. Optimization strategies**

Centralized PKI: provides centralized certificate issuance and management.

Automation: automatically issue and renew certificates.

Certificate monitoring: generate alerts before certificates expire.

Short-lived certificates: reduce dependence on long-term certificates but require reliable automation.

Automated revocation: immediately revoke certificates associated with compromised accounts.

Centralized policy management: enforce consistent security policies.

Conclusion: Automation and centralized PKI are essential for efficiently managing certificates at this scale.

## **5. Email Encryption Latency in Cloud-Based Systems**

Given: AES encryption = 0.2 ms/MB; RSA key exchange = 5 ms/email; attachment = 8 MB; emails = 8,000/hour.

### **A. Total encryption time per email**

AES time =  $0.2 \times 8 = 1.6$  ms.

RSA time = 5 ms.

Total encryption time =  $1.6 + 5 = 6.6$  ms/email.

Answer: 6.6 ms/email.

### **B. Total processing time per hour**

$6.6 \text{ ms} \times 8,000 = 52,800 \text{ ms} = 52.8$  seconds.

Answer: 52.8 seconds/hour of sequential-equivalent processing time. With concurrent processing, many emails can be encrypted simultaneously.

### **C. Performance bottleneck**

AES consumes 1.6 ms/email, while RSA consumes 5 ms/email.

RSA share =  $(5 / 6.6) \times 100 = 75.76\%$ .

AES share =  $(1.6 / 6.6) \times 100 = 24.24\%$ .

Answer: RSA key exchange is the major performance bottleneck.

### **D. Optimization strategies**

ECC adoption: elliptic-curve cryptography can provide comparable security with smaller keys and generally lower computational and network overhead.

Parallel processing: encrypt multiple emails concurrently.

Hardware acceleration: use cryptographic acceleration for encryption operations.

Connection/session reuse: avoid unnecessary repeated key-establishment operations where the protocol permits.

Efficient key management: reduce unnecessary asymmetric operations.

Hybrid encryption: use asymmetric cryptography for key exchange and AES for the actual email data.



Conclusion: Since RSA contributes about 75.76% of the encryption time, optimizing the asymmetric-key operation provides the greatest performance improvement.